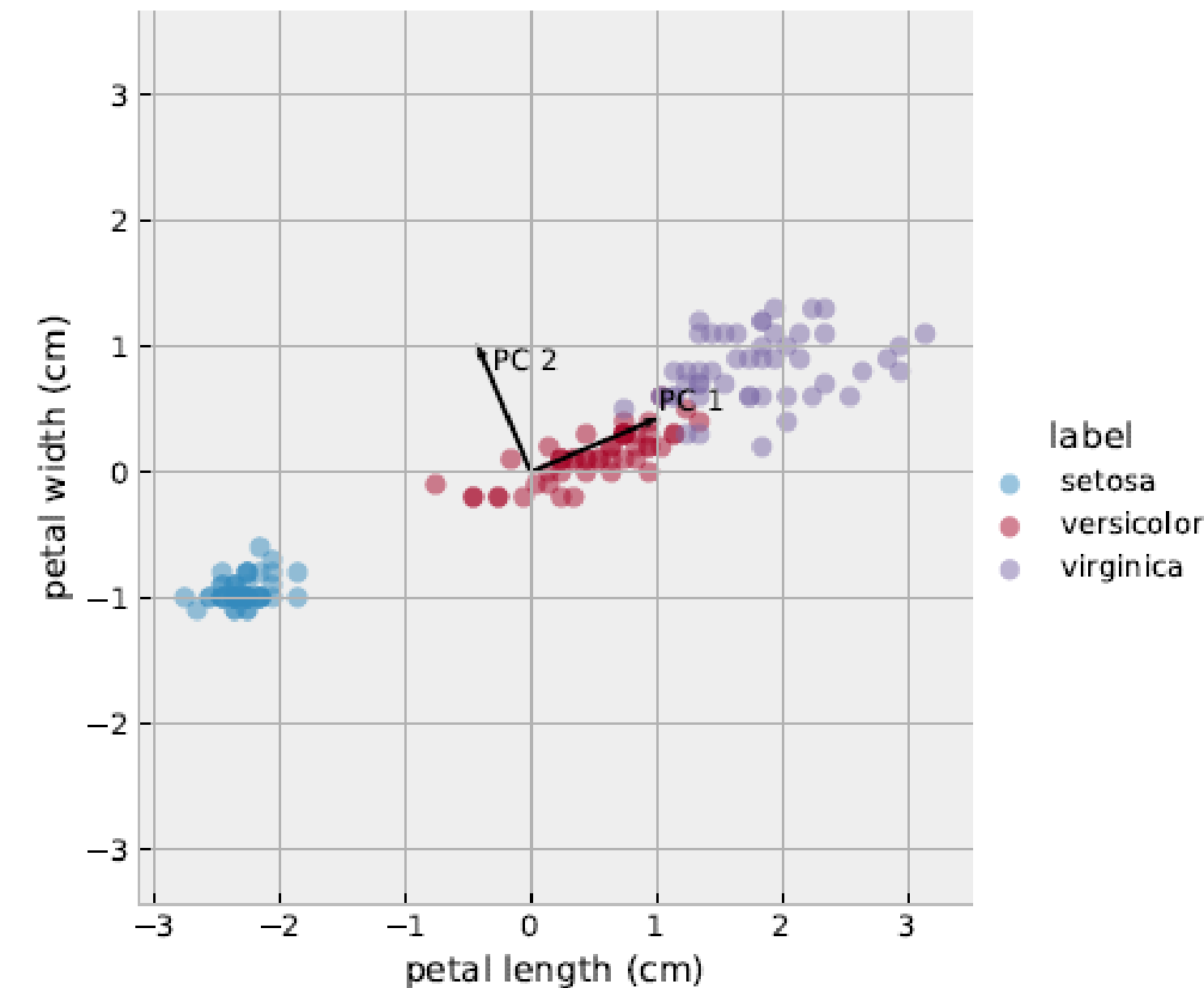
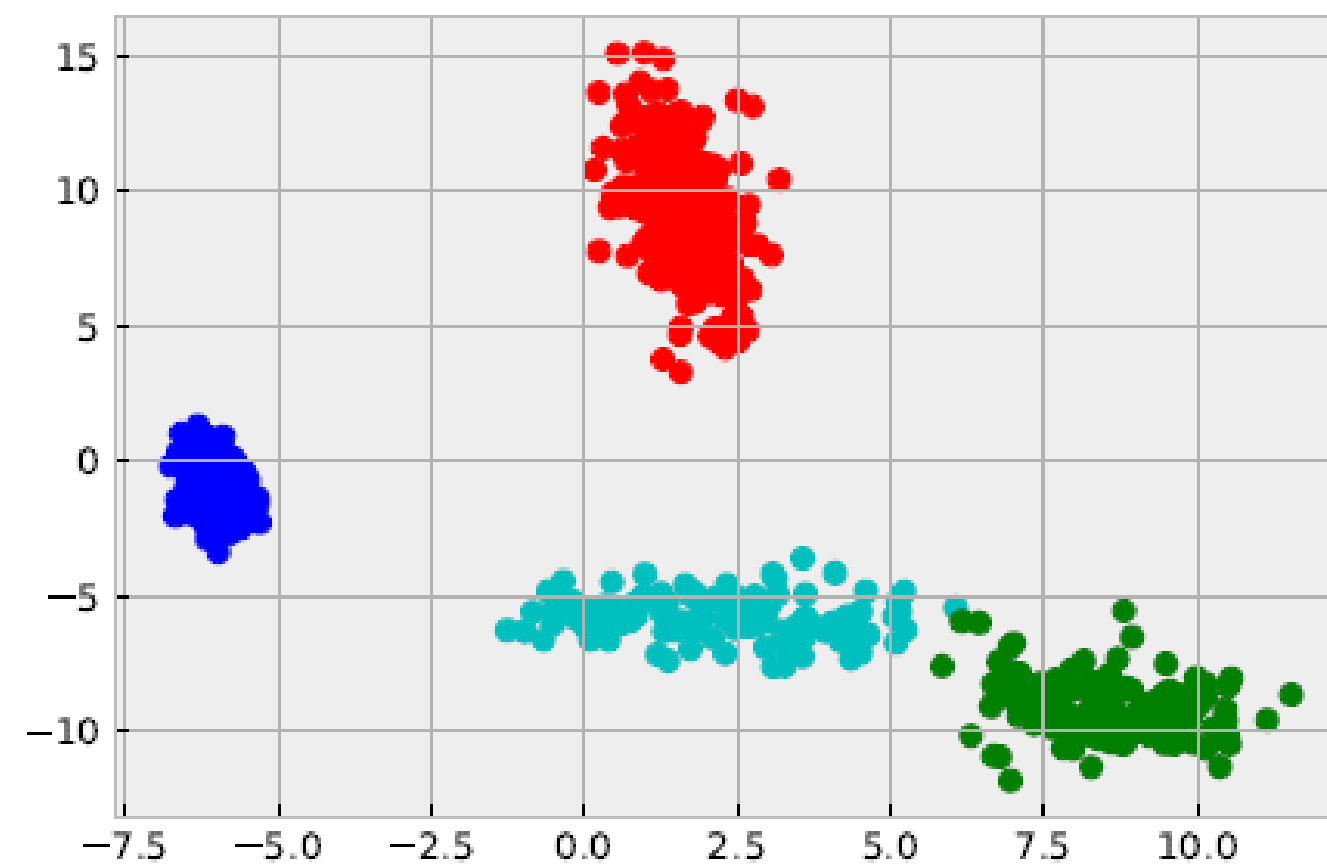


Applied Machine Learning

Dimensionality Reduction & PCA

2024-04-25

Main Use Cases for Unsupervised Learning



- Discover structure in data for interpretation
- Learn representations that make life easier for supervised or semisupervised learning

Why Feature Extraction?

- We have too many observations and dimensions
 - Need to “reduce” them to a smaller set of factors
 - To visualize
 - Too much noise in the data
 - Better representation of data without losing much information
 - Can build more effective data analyses on the reduced-dimensional space: classification, clustering, pattern recognition
- Combinations of observed variables may be more effective bases for insights, even if physical meaning is obscure

Basic Concepts

- Areas of variance in data are where items can be best discriminated and key underlying phenomena observed
- If two items or dimensions are highly correlated or dependent
 - They are likely to represent highly related phenomena
 - If they tell us about the same underlying variance in the data, combining them to form a single measure is reasonable
- So we want to combine related variables, and focus on uncorrelated or independent ones, especially those along which the observations have high variance
- We want a smaller set of variables that explain most of the variance in the original data, in more compact and insightful form

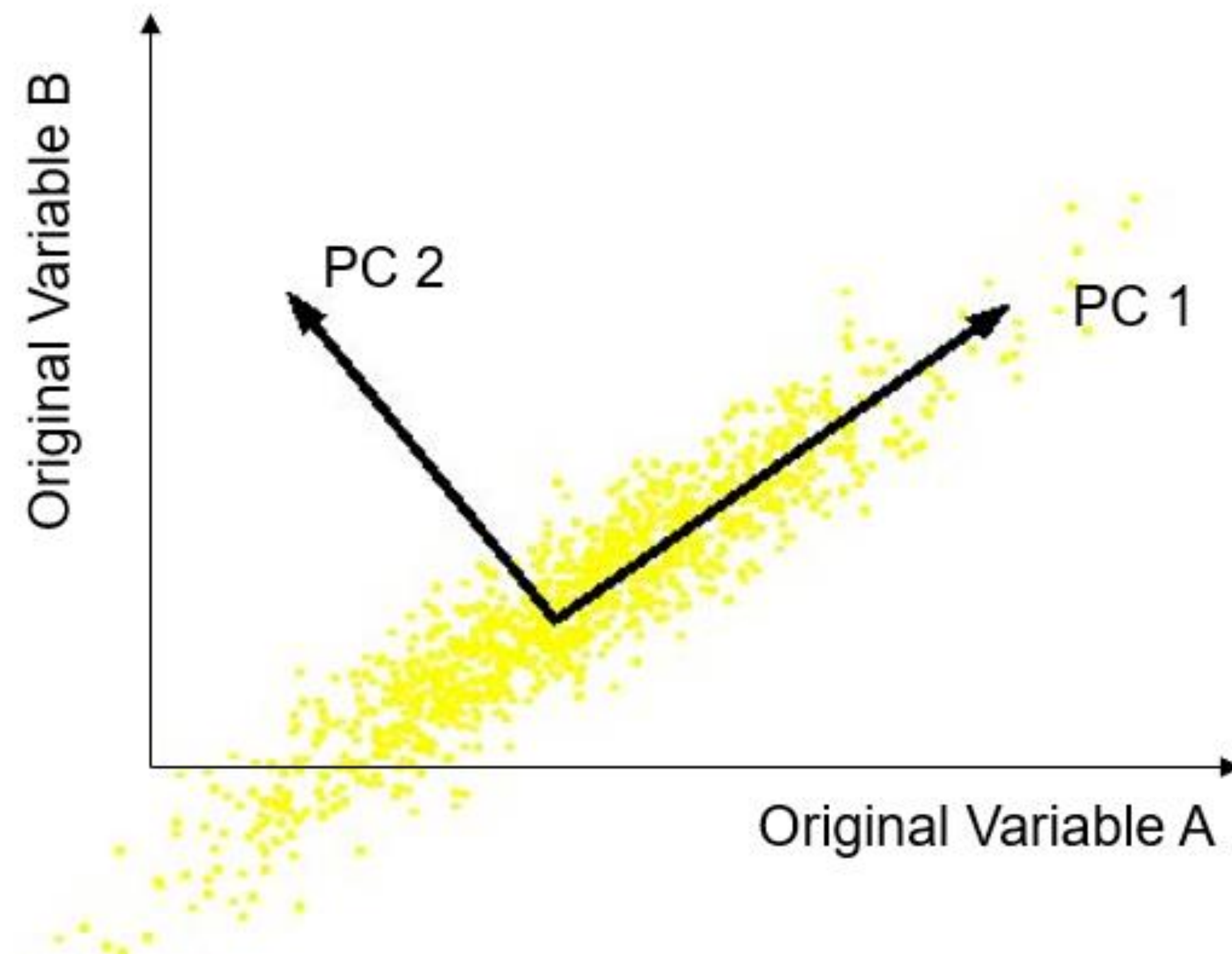
Basic Concepts

- What if the dependences and correlations are not so strong or direct?
- And suppose you have 3 variables, or 4, or 5, or 10000?
- Look for the phenomena underlying the observed covariance/co-dependence in a set of variables
- These phenomena are called “factors” or “principal components” or “independent components,” depending on the methods used
 - Factor analysis: based on variance/covariance/correlation
 - Independent Component Analysis: based on independence

Principal Component Analysis (PCA)

- Most common form of factor analysis
- The new variables/dimensions
 - Are linear combinations of the original ones
 - Are uncorrelated with one another
 - Orthogonal in original dimension space
 - Capture as much of the original variance in the data as possible
 - Are called Principal Components

Principal Component Analysis (PCA)



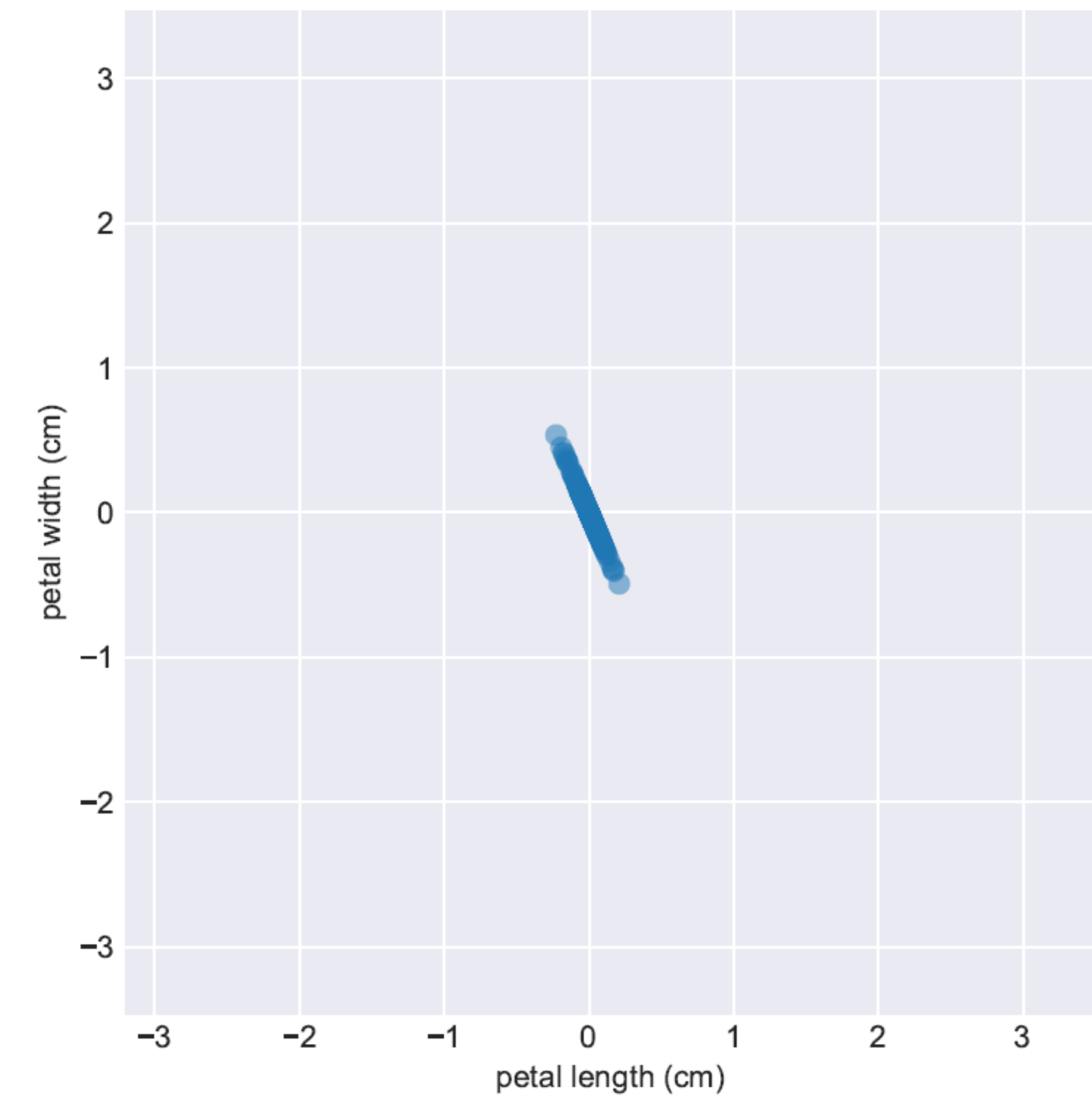
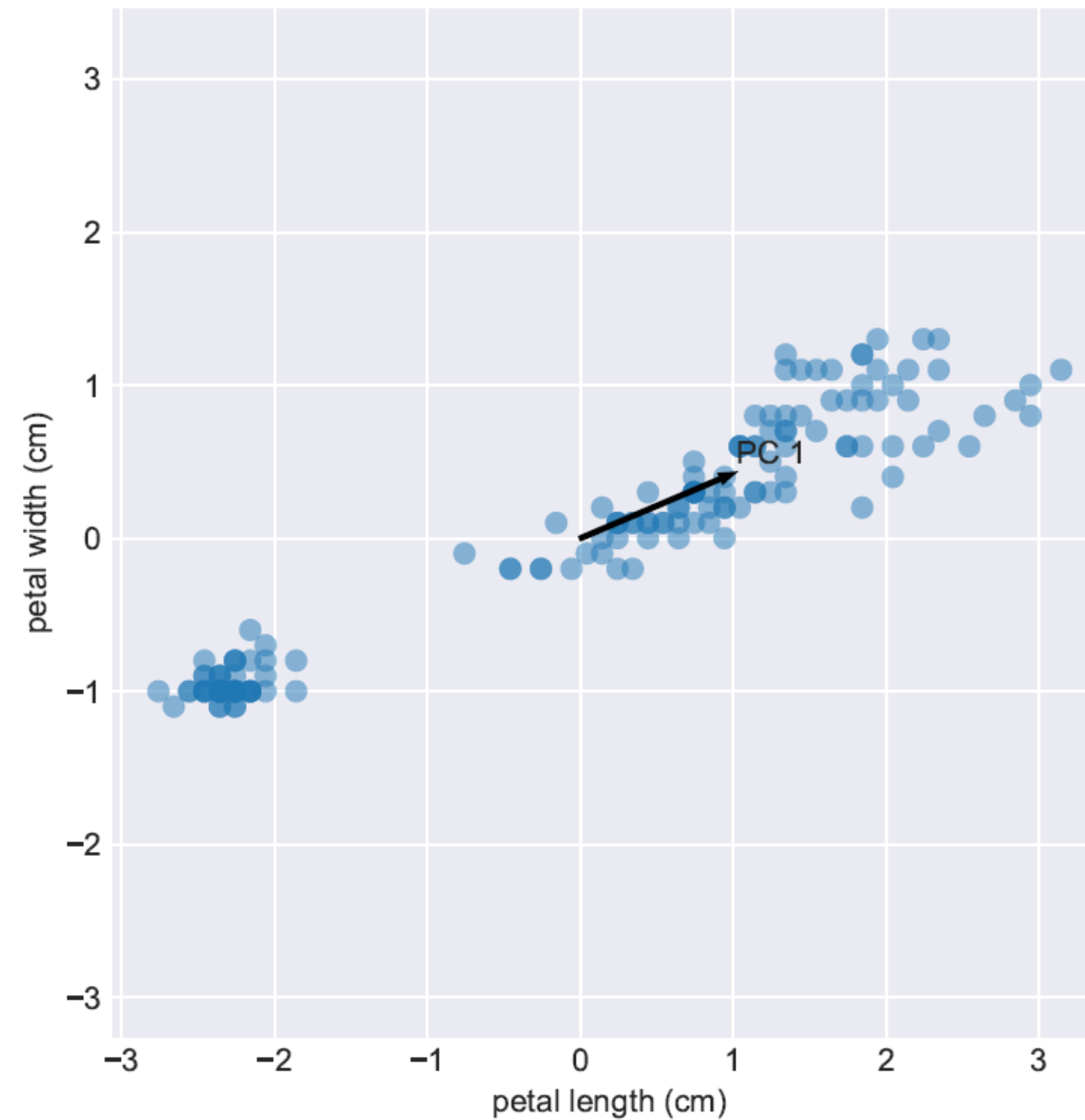
- Orthogonal directions of greatest variance in data
- Projections along PC1 discriminate the data most along any one axis

Principal Component Analysis (PCA)

- **Principal components:**

- First principal component is the direction of greatest variability (covariance) in the data
- Second is the next orthogonal (uncorrelated) direction of greatest variability
- So first remove all the variability along the first component, and then find the next direction of greatest variability
- And so on ...

Principal Component Analysis (PCA)



- Example: Iris data (after removing the first principal component)

Principal Component Analysis (PCA)

- **Principle**

- Linear projection method to reduce the number of parameters
- Transfer a set of correlated variables into a new set of uncorrelated variables
- Map the data into a space of lower dimensionality
- Form of unsupervised learning

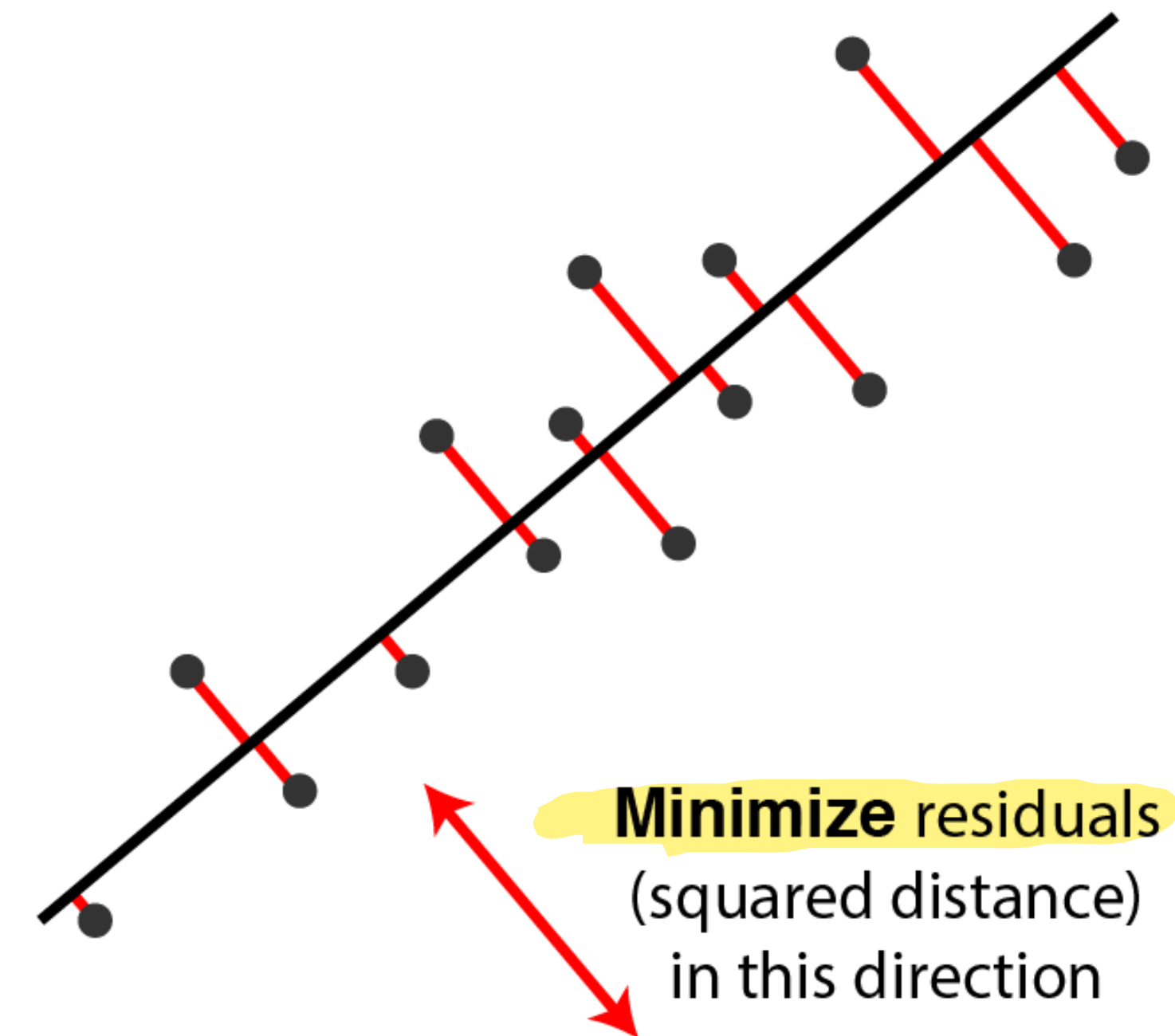
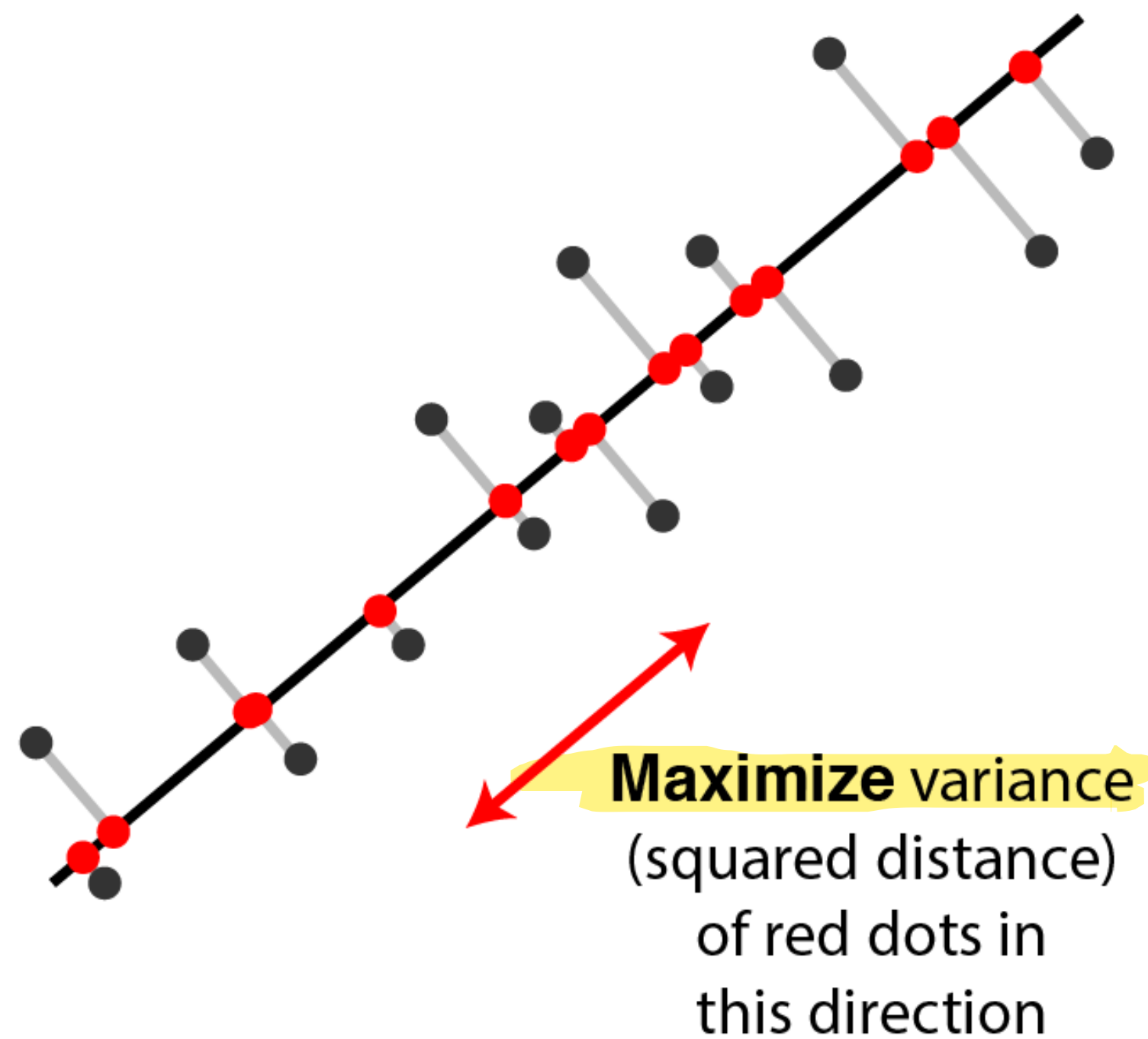
- **Properties**

- It can be viewed as a rotation of the existing axes to new positions in the space defined by original variables
- New axes are orthogonal and represent the directions with maximum variability

Principal Component Analysis (PCA)

- Data points are vectors in a multidimensional space
- Projection of vector $\mathbf{x}$ onto an axis (dimension) $\mathbf{u}$ is $\mathbf{u} \cdot \mathbf{x}$
- Direction of greatest variability is that in which the average square of the projection is greatest
 - i.e. $\mathbf{u}$ such that $\mathbf{E}((\mathbf{u} \cdot \mathbf{x})^2)$ over all $\mathbf{x}$ is maximized
 - we subtract the mean along each dimension, and center the original axis system at the centroid of all data points, for simplicity
 - This direction of $\mathbf{u}$ is the direction of the first Principal Component (PC1)

Principal Component Analysis (PCA)



Principal Component Analysis (PCA)

- $E((\mathbf{u} \cdot \mathbf{x})^2) = E((\mathbf{u} \cdot \mathbf{x})(\mathbf{u} \cdot \mathbf{x})^T) = E(\mathbf{u} \cdot \mathbf{x} \cdot \mathbf{x}^T \cdot \mathbf{u}^T)$
- The matrix $\mathbf{C} = \mathbf{x} \cdot \mathbf{x}^T$ contains the correlations (similarities) of the original axes based on how the data values project onto them
- So, we are looking for $\mathbf{u}$ that maximizes $\mathbf{u} \mathbf{C} \mathbf{u}^T$, subject to $\mathbf{u}$ being unit-length
- It is maximized when $\mathbf{u}$ is the principal eigenvector of the matrix $\mathbf{C}$, in which case
 - $\mathbf{u} \mathbf{C} \mathbf{u}^T = \mathbf{u} \boldsymbol{\lambda} \mathbf{u}^T = \boldsymbol{\lambda}$ if $\mathbf{u}$ is unit-length, where $\boldsymbol{\lambda}$ is the principal eigenvalue of the correlation matrix $\mathbf{C}$
 - The eigenvalue denotes the amount of variability captured along that dimension

Principal Component Analysis (PCA)

- Maximise $\mathbf{u}^T \mathbf{X} \mathbf{X}^T \mathbf{u}$ s.t $\mathbf{u}^T \mathbf{u} = 1$
- Construct Lagrangian $\mathbf{u}^T \mathbf{X} \mathbf{X}^T \mathbf{u} - \lambda \mathbf{u}^T \mathbf{u}$
- Vector of partial derivatives set to zero:

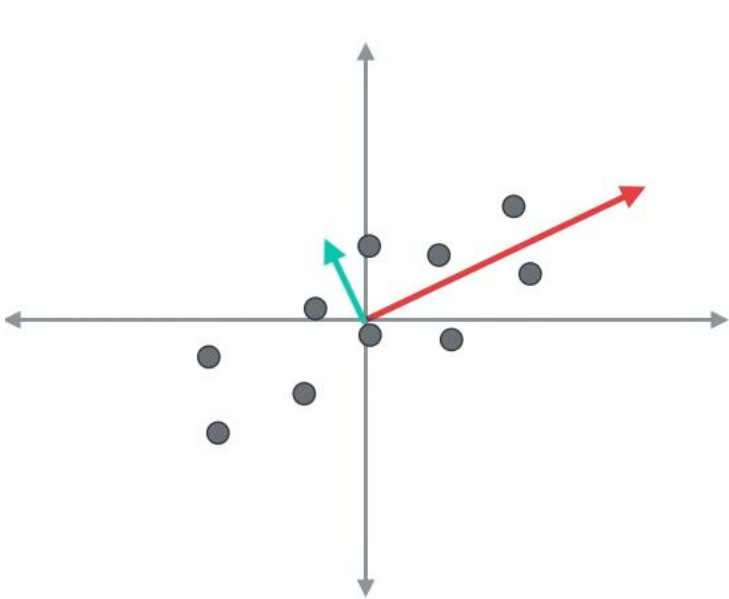
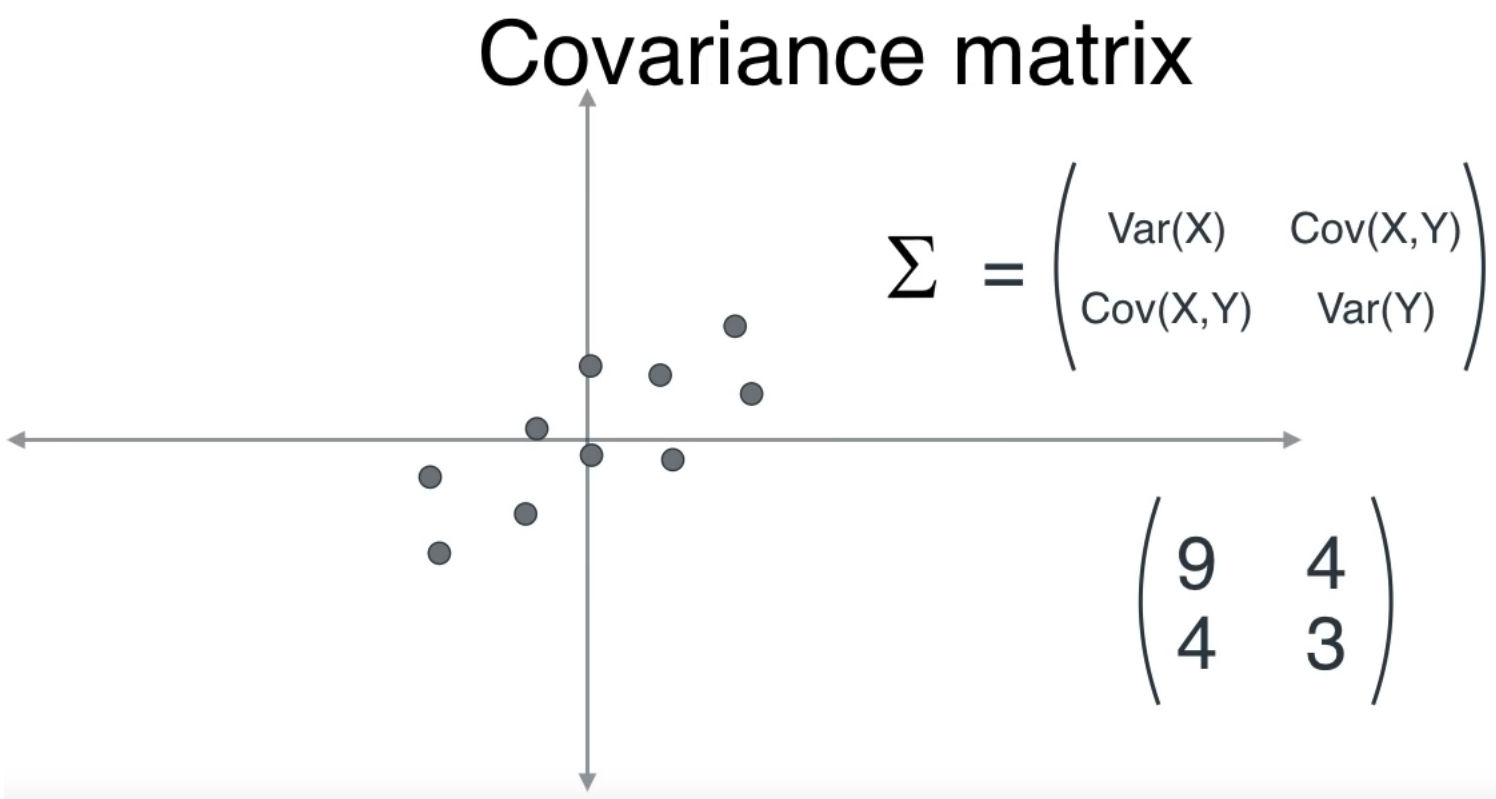
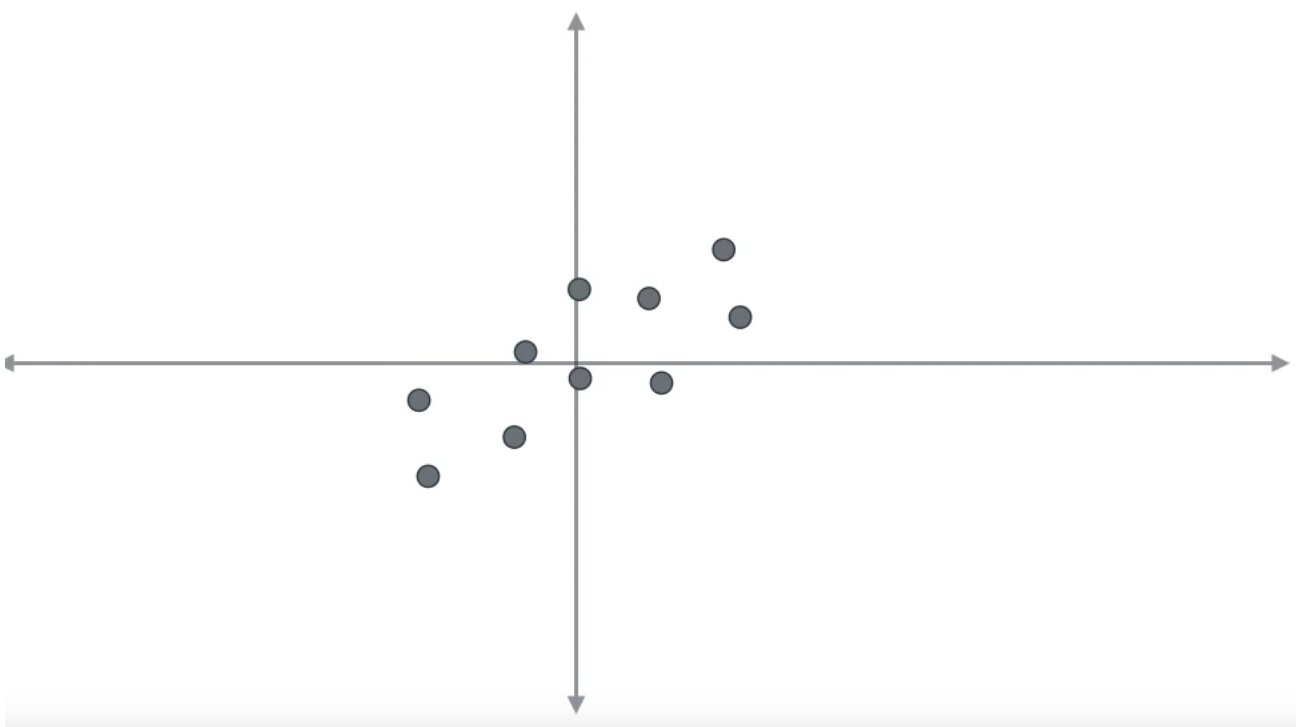
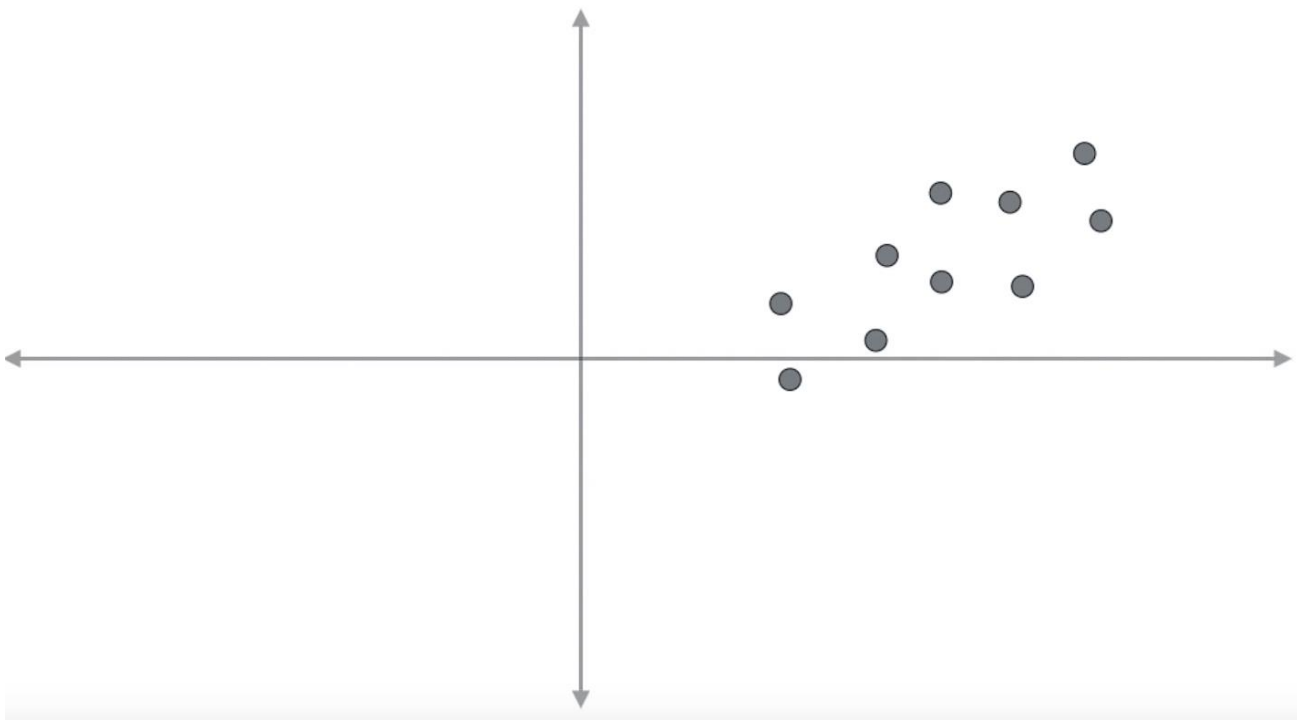
$$\mathbf{X} \mathbf{X}^T \mathbf{u} - \lambda \mathbf{u} = (\mathbf{X} \mathbf{X}^T - \lambda \mathbf{I}) \mathbf{u} = \mathbf{0}$$

- As $\mathbf{u} \neq \mathbf{0}$ then $\mathbf{u}$ must be an eigenvector of $\mathbf{X} \mathbf{X}^T$ with eigenvalue λ

Principal Component Analysis (PCA)

- A linear dimensionality reduction method.
- The components of PCA are linearly uncorrelated with each other. The components are ordered based on the variance covered by the components; this means that the first component (PC1) explains the most variance in the data.
- To find PC1, we look for a direction vector with unit length (vector length is 1) in which the variance of X (the data) is maximal.
- We can use SVD to implement PCA.
- When input features are on very different scales, apply feature scaling before applying PCA.

Principal Component Analysis (PCA)

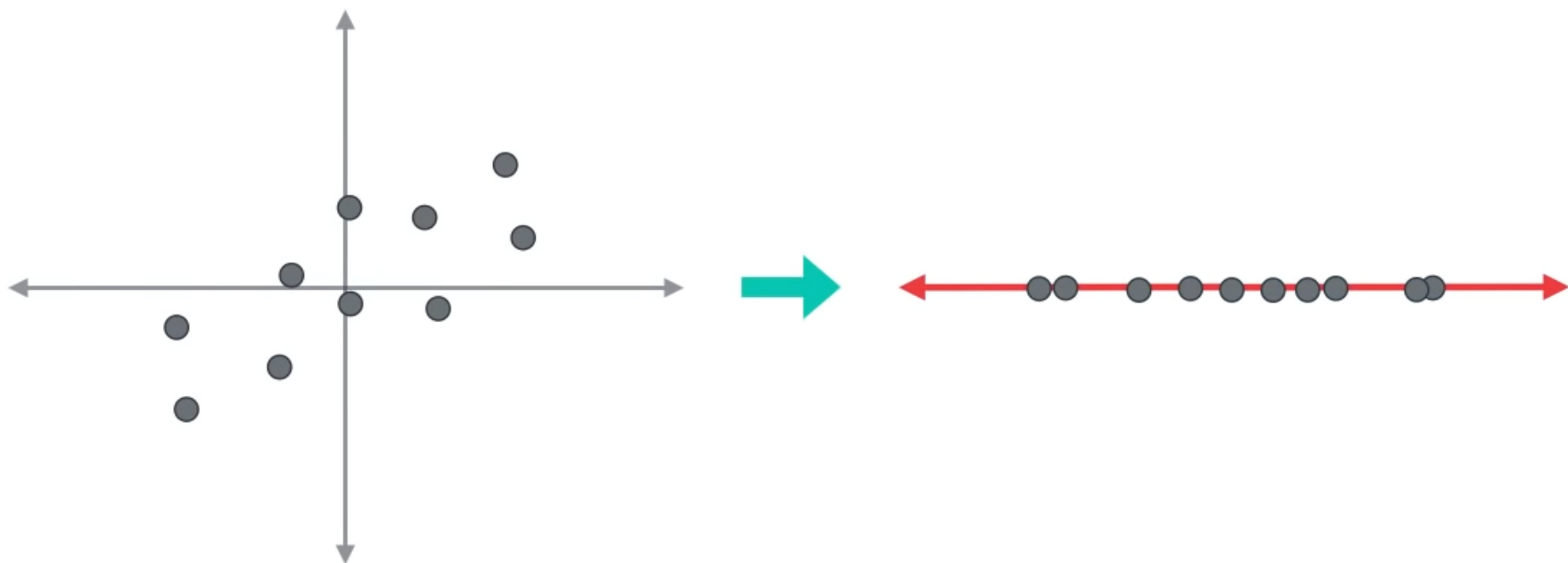
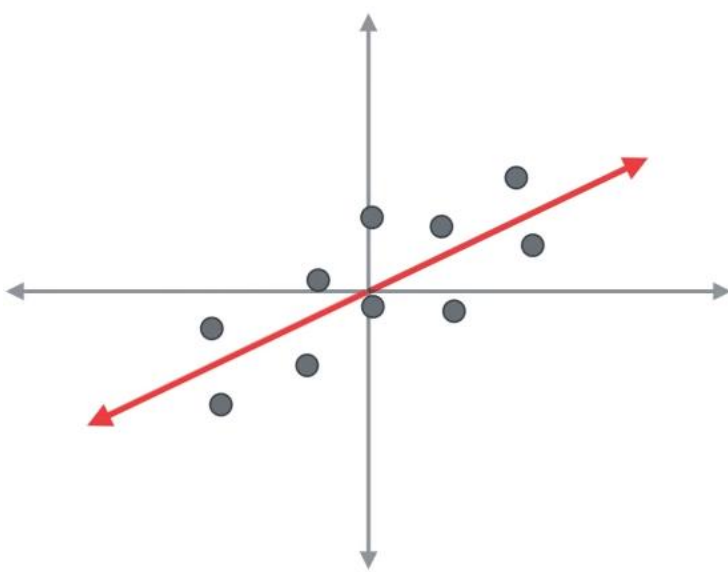


$$\Sigma = \begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$
$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Eigenvectors
(direction)

$$11 \quad 1$$

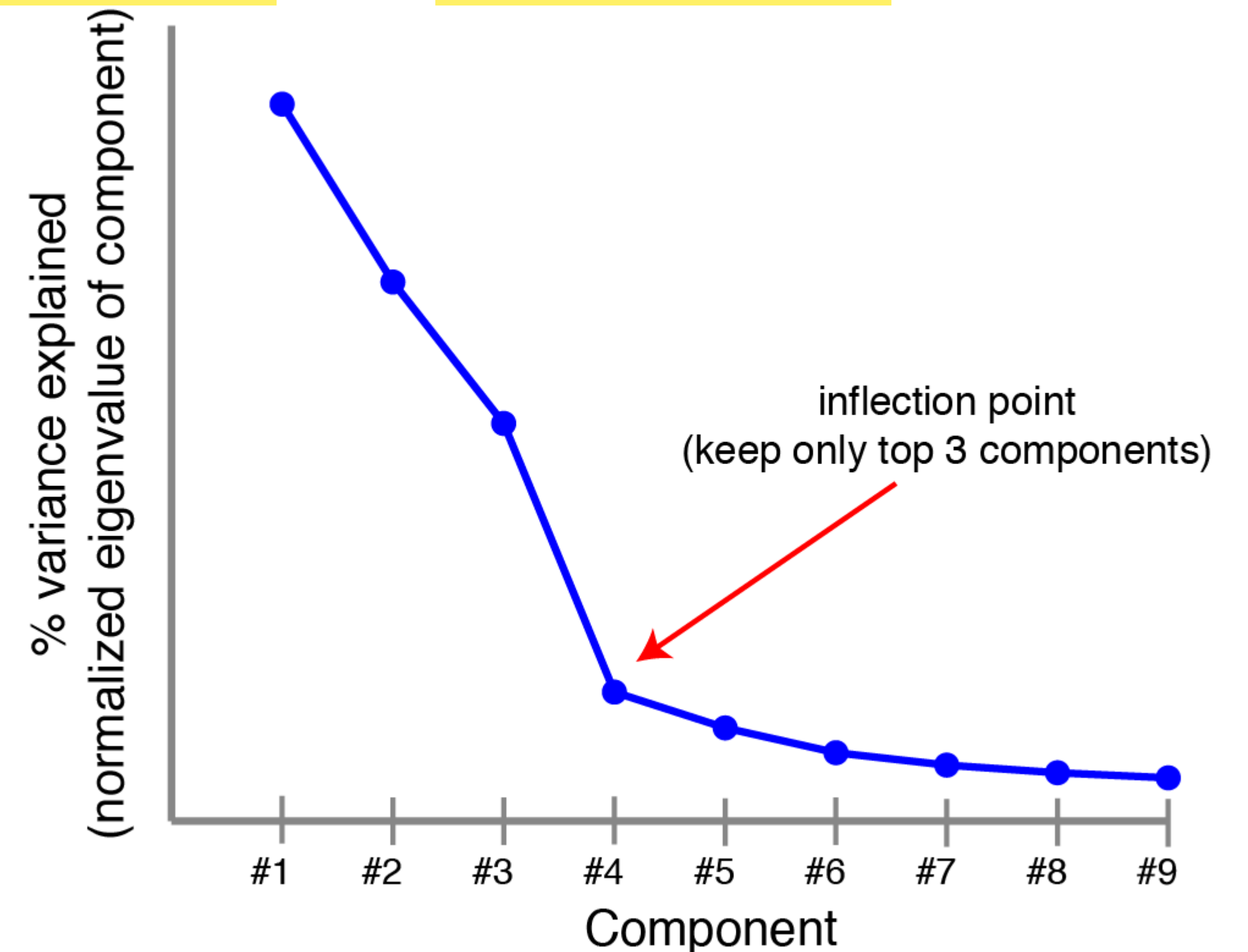
Eigenvalues
(magnitude)



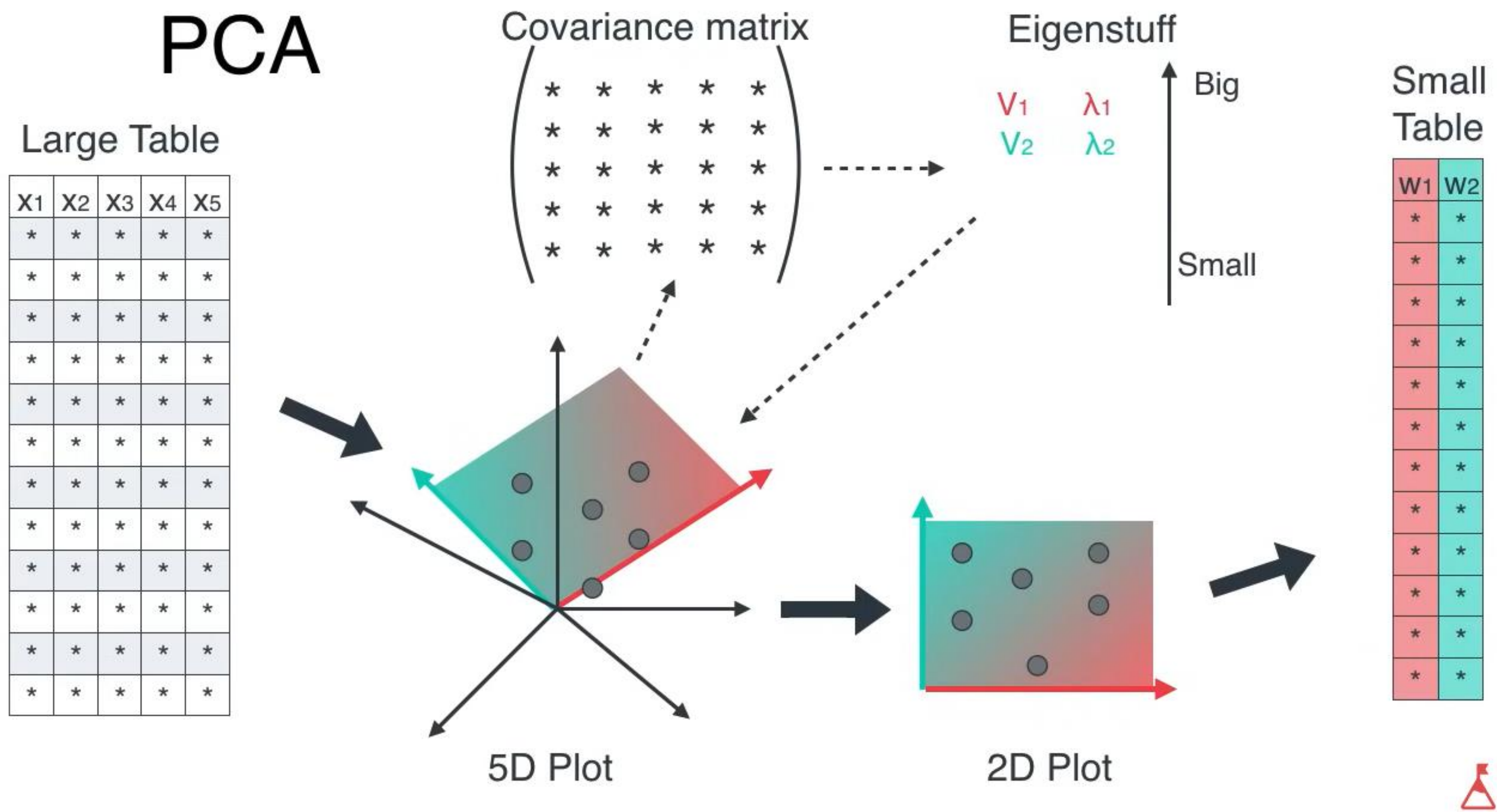
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Principal Component Analysis (PCA)

- Can ignore the components of lesser significance.
- You do lose some information, but if the eigenvalues are small, you don't lose much
 - n dimensions in original data
 - calculate n eigenvectors and eigenvalues
 - choose only the first p eigenvectors, based on their eigenvalues
 - final data set has only p dimensions

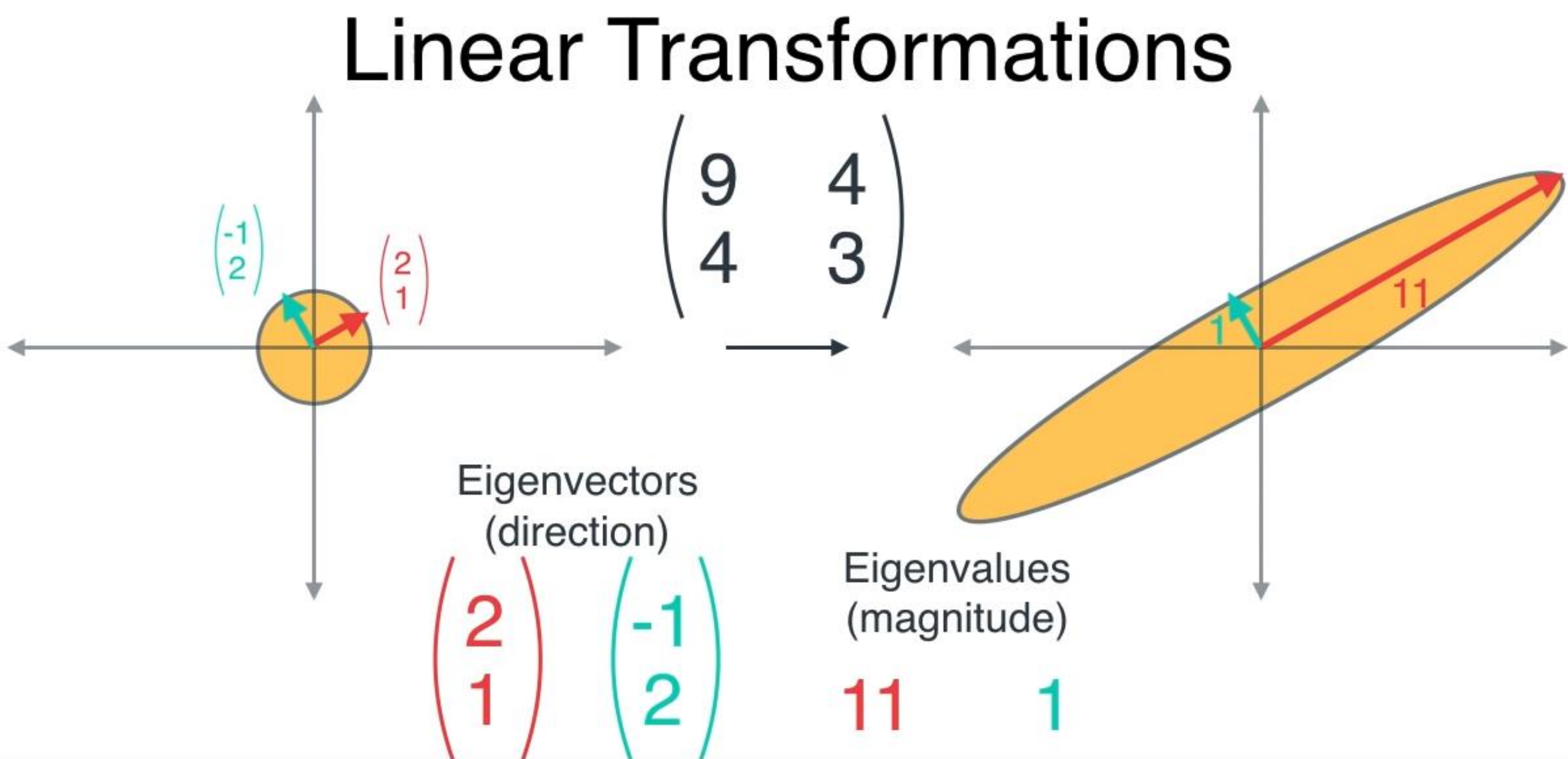
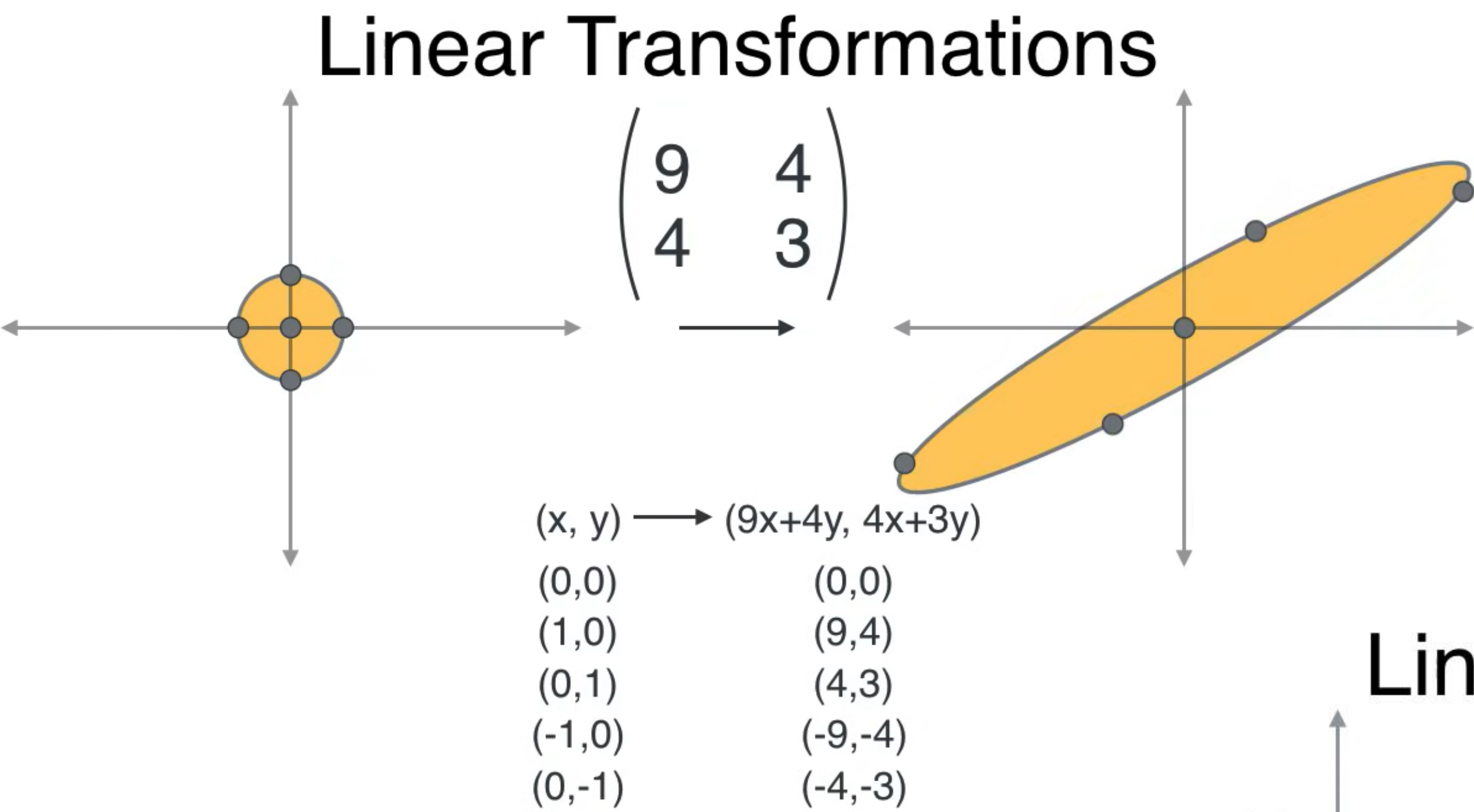


Principal Component Analysis (PCA)

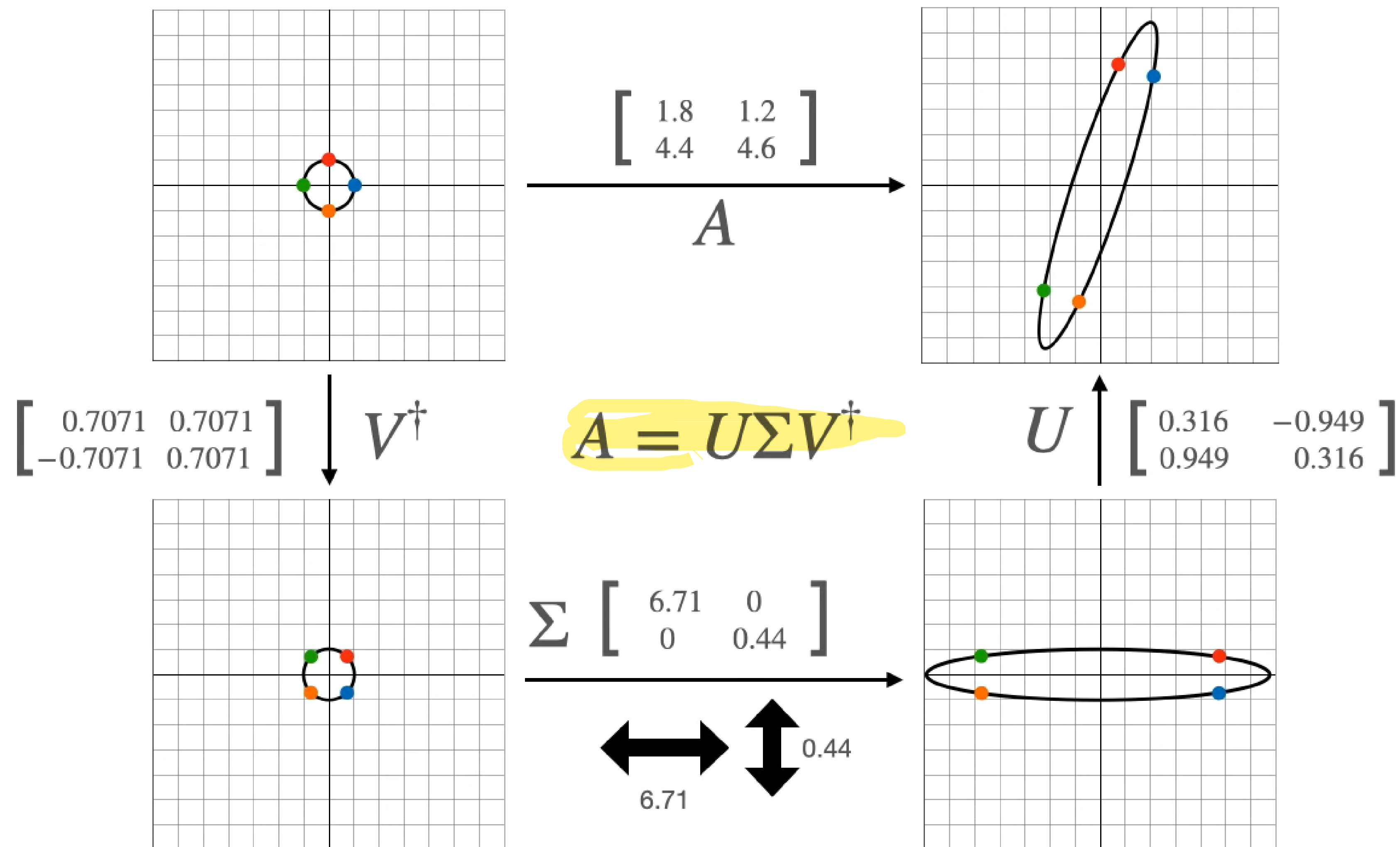


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Linear Transformations



Singular Value Decomposition (SVD)



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Singular Value Decomposition (SVD)

Rank 1 matrices

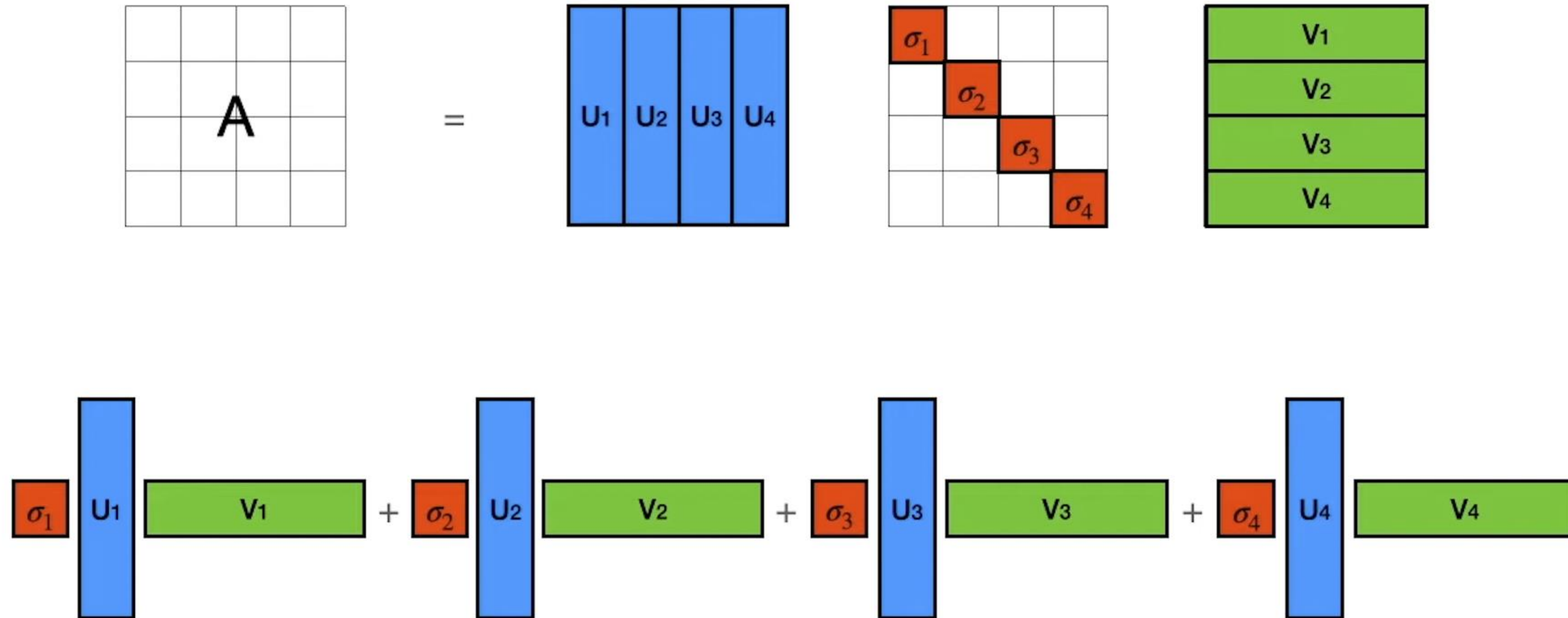
1	2	3	4
-1	-2	-3	-4
2	4	6	8
10	20	30	40

 $=$

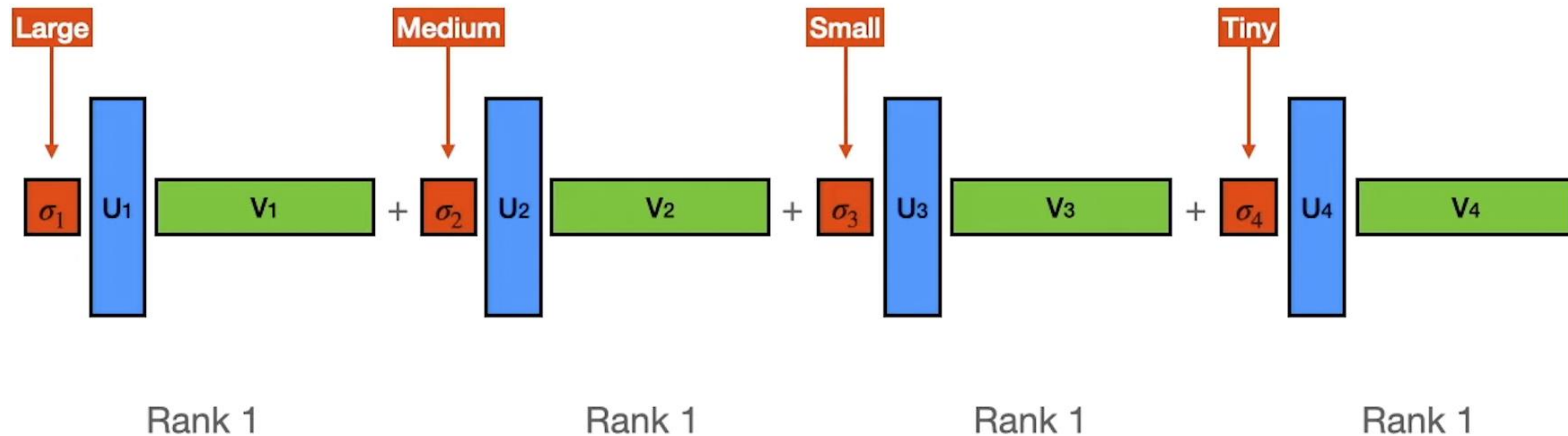
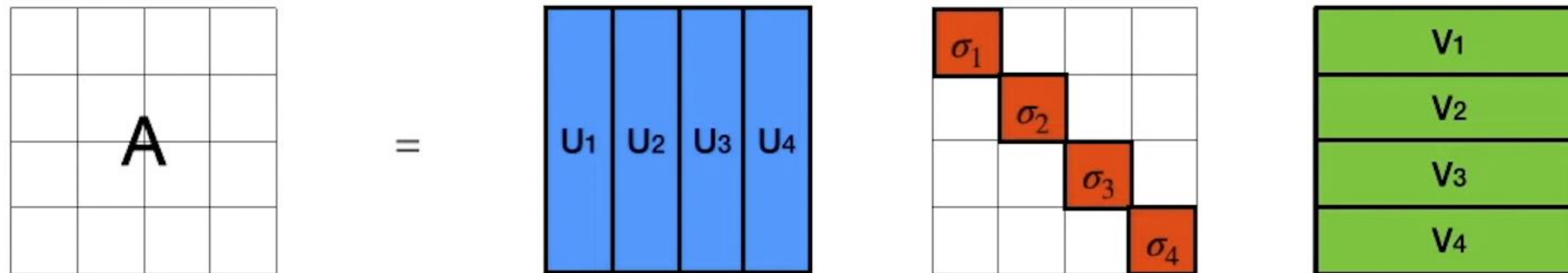
1
-1
2
10

1	2	3	4
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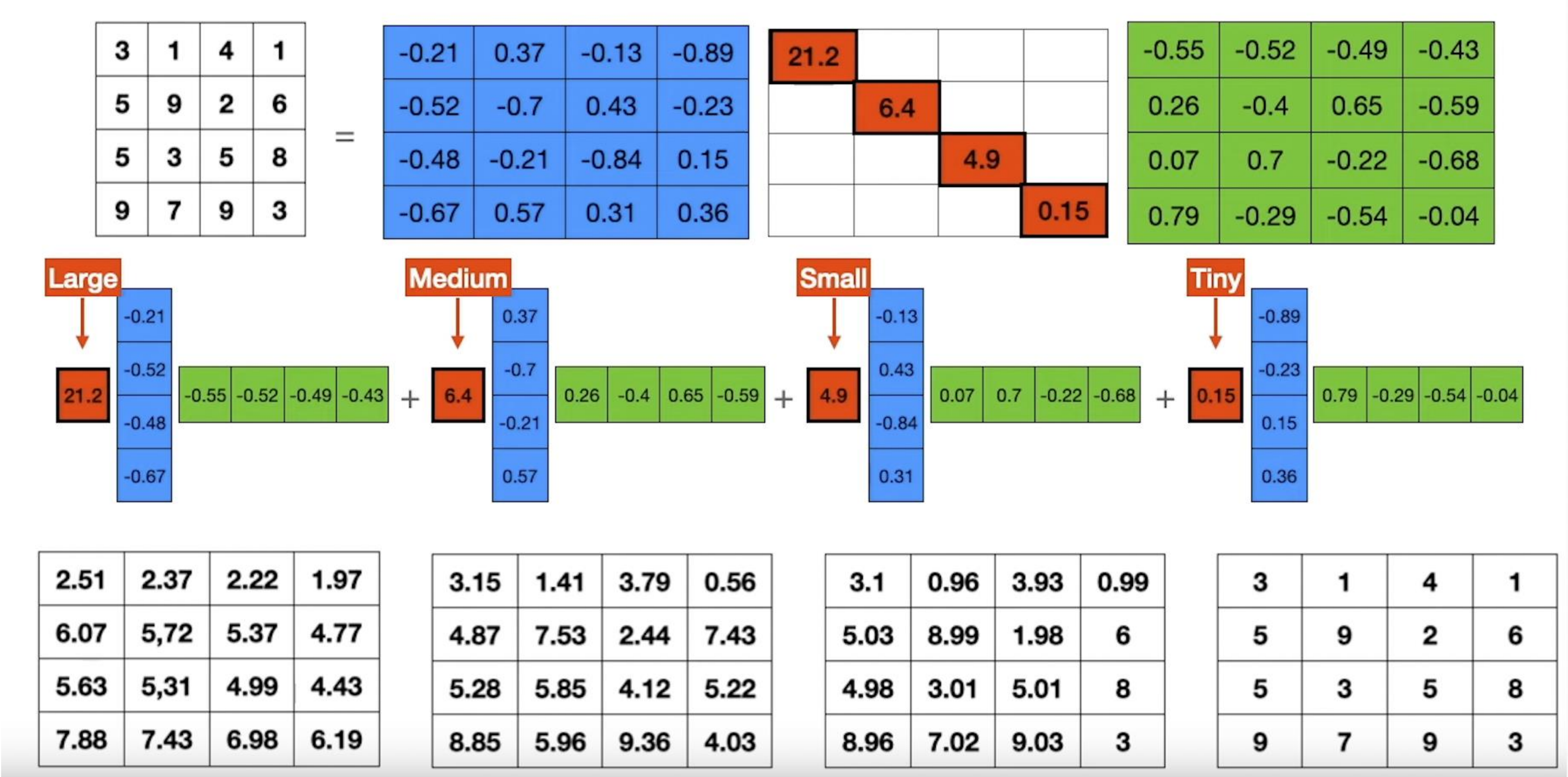
Singular Value Decomposition (SVD)



Singular Value Decomposition (SVD)



Singular Value Decomposition (SVD)



[Source](#)

Principal Component Analysis (PCA)

- Videos describing PCA ideas in simpler way:
[Basic idea](#)
- Video describing practical tips for PCA:
[practical tips](#)
- Video describing Singular Value Decomposition (SVD):
[SVD](#)
- Video describing Eigenvectors and Eigenvalues:
[Eigenvectors](#)